

Embedded Systems & IoT April -July 2026

Lab Exercise: Schematics with ESP32

Instructions:

For this exercise, you will largely use the EasyEDA software for circuit design. In part A, you can write the results of your calculations to a suitable document and export it as a PDF. For part B, export the designs directly from EasyEDA as PDF documents.

Part A

166985, Parneet Kaur 4.1E

1) calculate the total resistance in ohms.

Circuit a (series)

$$R = R_1 + R_2 + R_3$$

$$= 20\Omega + 30\Omega + 50\Omega$$

$$= 100\Omega$$

Circuit b (parallel)

$$= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_T} = \frac{1}{20\Omega} + \frac{1}{100\Omega} + \frac{1}{50\Omega} = \frac{7}{25\Omega}$$

$$R_T = \frac{1}{\frac{7}{25\Omega}} = 12.5\Omega$$

2) calculate the total current from the power supply.

Circuit a (series)

$$I = \frac{V}{R} = \frac{125}{100} = 1.25A$$

Circuit b (parallel)

$$I = \frac{V}{R} = \frac{125}{12.5} = 10A$$

3) calculate the current through each resistor.

Circuit a (series)

$$I_1 = I_2 = I_3 = 1.25A$$

current is same everywhere.

Circuit b (parallel)

voltage is same across each resistor (125V)

for 20Ω $I = \frac{125}{20} = 6.25A$

100Ω $I = \frac{125}{100} = 1.25A$

50Ω $I = \frac{125}{50} = 2.5A$

$$6.25 + 1.25 + 2.5 = 10A$$

4) calculate the voltage drop through each resistor.

Circuit a (series)

$$V = IR$$

20Ω : $V = 1.25 \times 20$

$$V = 25V$$

30Ω : $V = 1.25 \times 30$

$$= 37.5V$$

50Ω : $V = 1.25 \times 50$

$$= 62.5V$$

$$= 25 + 37.5 + 62.5 = 125V$$

Circuit b (parallel)

voltage is same everywhere

$$V_{20\Omega} = V_{100\Omega} = V_{50\Omega} = 125V$$

5) calculate the power dissipated through each resistor. (use watt's law : Power = voltage x current)

circuit a:

$$20\Omega : P = VI$$

$$= 25 \times 1.25 = 31.25W$$

$$30\Omega = P = 37.5 \times 1.25 = 46.875W$$

$$50\Omega = P = 62.5 \times 1.25 = 78.125W$$

circuit b

$$20\Omega : P = 125 \times 6.25 = 781.25W$$

$$300\Omega : P = 125 \times 1.25 = 156.25W$$

$$50\Omega : P = 125 \times 50 = 312.5W.$$